Electrocardiogram lab

Prelims

This lab involves taking ECG measurements and writing a report. You only need to turn in one report per group. Remember that your ECG is medically-privileged information, and you do not need to collect it or share it with others. While part of the fun (for many people) is hooking up your body to your equipment and learning about your own ECG, if you prefer you can either turn in the ECG of another person in your group, or use the sample ECG that is provided with this lab on the course web page.

As usual, you only need to turn in one report. But feel free to have fun and each collect your own data and compare the readings with each other. If there are substantial discrepancies, feel free to put the graphs for multiple people in the same report (you don't have to tell me whose pictures are whose unless you want to).

ECG terminology: electrodes and leads

Each wire that you connect to your body is called an *electrode*. Basic physics says that voltage is defined as the difference between the electric potential at two different points, and so any particular reading that you see on a screen is the voltage difference between two electrodes. Each of these readings is called a *lead*. I don't know why it's called a lead – but that's standard medical terminology, even if it usually sounds a bit strange to engineers at first.

A typical ECG in a doctor's office attaches 10 electrodes to your body and displays 12 leads. With 10 electrodes, it would only take 9 leads to fully represent the data, and so we have some redundant readings – the redundant readings make it easier for a clinician to quickly diagnose issues.

In this lab, we will attach four electrodes and use them to collect six leads (I, II, III, aV_L, aV_R and aV_F).

Noise in ECGs

The signals that your heart produces are quite strong enough (> 60mV) in your heart, which is plenty to travel to the next cardiomyocyte and cause a contraction. By the time they travel through your body to our electrodes, though, the signal is only a few mV. It is easy for environmental noise to overwhelm an ECG, and the most common source of environmental noise is the 60Hz noise from power distribution and appliances in homes and buildings.

The ECG amplifier is a *differential amplifier*; if the same noise is applied to both electrodes of any lead, then the lead (which measures the difference between its two electrodes) will not show the noise. Thus, the main source of noise that we see in our readouts is noise that affects the two electrodes differently.

However, the differential amplifier is not perfect; it only works well if both electrodes average roughly zero volts. The purpose of the red electrode is to bring the other electrodes down to an average of 0V, and thus minimize noise. It uses a system called *right-leg drive* (RLD), where the AD8232 board tries to dynamically find noise that is common to both electrodes and then drive it back to your body (via your right leg) to cancel it out. In fact,

static electricity can cause your body to float 100V or more above ground; RLD smoothly brings it back near zero.

Hooking up the pads to your body

The board has two earphone jacks, labeled Channel 1 and Channel 2. Each should get a set of electrodes plugged in, giving us 2*3=6 electrodes (of which we will only use 4). The two channels are *not* interchangeable.

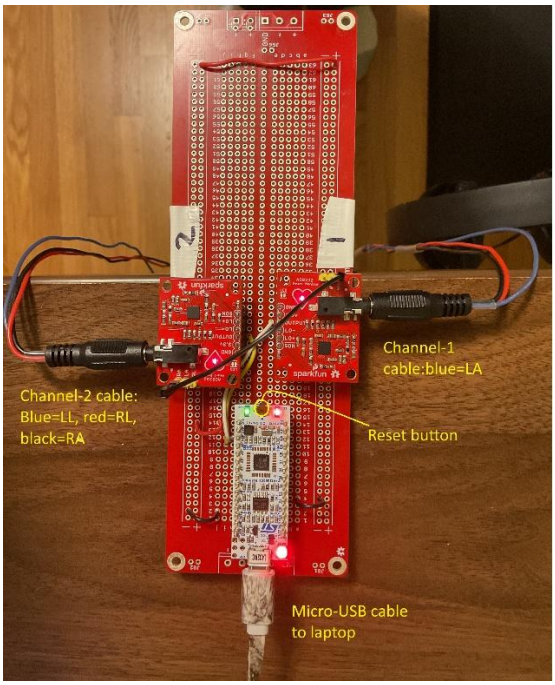
- Channel 1: the blue electrode connects to your left arm (LA), and the red and black electrodes are unused.
- Channel 2: the blue electrode connects to your left leg (LL); the red electrode to your right leg (for RLD), and the black electrode to your right arm (RA)
- The red RLD electrode should always be attached to your right leg.

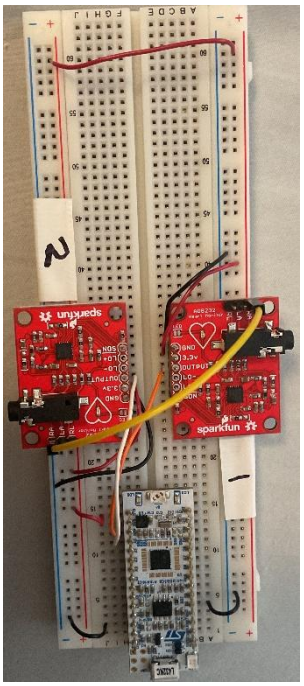
There is a video showing the placement on the course web page; the link is with the rest of the information for this lab.

Collecting the data

Collecting the data is the hard part of this lab.

First, grab one of the boards. Your board may look like the one on the left or like the one on the right; either one is fine. (None of the boards come with the cables already attached).





Attach it to your laptop with one of the USB cables provided.

Next, download the Python script *scope_host.py* from the course web site. Load it into your favorite Python environment (e.g., Thonny), but don't run it yet.

Grab two sets of electrodes. Once you have them, attach them to the pads on your body as described above. Plug the "headphone" plugs on the wires into the matching jacks on the board.

You're ready to go!

Run *scope_host.py*. If it doesn't immediately display the ECG traces for lead 1 and lead 2, then try pressing the reset button on the board. When you do have a nice trace showing, then type an "r" (this will only work if your mouse focus is on the plot window, not the Python-IDE window). The "r" will trigger 10 seconds of data collection; when the data collection is done, it will create the file *scope_out.csv* (most likely in whatever folder you downloaded *scope_host.py* into).

If the plotting window either does not pop up or does not display a graph of your ECG, you may have to install an *ST-link driver*. You can find it at https://www.st.com/en/development-tools/stsw-link009.html . Install it on your laptop. This enables the data-capture boards to talk to your laptop (usually this isn't necessary).

Running the ECG-plotting tool

The ECG-plotting tool is called *6lead_plot.py*, and you can download it from the course website. Open it in your favorite Python environment as usual. At the top of the file, you will notice that two variables are defined:

```
ecg_filename = 'scope_out.csv'
PQRST = (1,1.23,1.45,1.67,2.11)  # Put your own 5 numbers here
```

The top line means that the plotting tool will read the file *scope_out.csv*, which is the file that the data-collection tool generated. Don't touch the *PQRST* numbers for now.

Next, run the program as usual. Hopefully, a plot of the six leads I, II, III, aV_L, aV_R and aV_F will appear. Zoom in/around the plot and locate the peak of the P, Q, R, S and T waves (it may be useful to hold down the "x" key when you zoom, which keeps the Y axis untouched). When you have located these points, now update the *PQRST* variable accordingly.

With this done, when you rerun the plotting program you will have all five waves marked on all six views. Capture the pictures to turn in.

Readings with muscular involvement

Our previous bodily-measurement lab read a surface EMG by using a very similar experimental setup to the one we're using today. So why does today's lab display our ECG signal and not an EMG? And why didn't our EMG lab display an ECG waveform? Let's run some experiments to better understand why.

Take your six-lead ECG as before, but this time clench your left biceps for the first few seconds of the recording; then relax your left arm and clench your right biceps for the next few seconds. Again, capture this waveform. For this set of readings, there's no need to find the PQRST points.

Note, by the way, that when you rerun the data-collection tool to capture your ECG with muscular involvement it will again save its output to *scope_out.csv*, which will overwrite your earlier data capture. Feel free to rename the earlier data if you don't want that.

Once you have a *scope_out.csv* for the muscular-involvement capture, use Excel or a similar tool to read it and plot the signals for Lead I (the first column) and Lead II (the second column).

Using prerecorded traces

As noted above, ECG traces are private medical information, and capturing your own ECG traces is optional. If you decide to use the traces available on the course web site, then notice that they will not be named *scope_out.csv*. You must thus either rename them, or change *6lead_plot.py* to read the appropriate .csv file.

Discussion questions:

1. Turn in your plots, both for the regular run (from *scope_host.py*) and the muscular-involvment runs (from Excel). For the regular run, please zoom in enough that we can tell reasonably easily if you've correctly found the peaks. (25 points for good pictures, and 25 for correctly finding the P,Q,R,S and T peaks).
2. Given your images, can you read the numbers for:
    a. Heartrate
    b. PR segment (i.e., the delay across the AV node)

    In each case, you should also annotate the plots to show where you're grabbing the numbers from. (10 points)
3. Based on the images, can you compute your R angle? The normal range for adults is typically given as -30º to +90º; are you within that range? (To make grading easier, please show your work using a clock-face circle as we did in class). Look up the terms "vertical" and "horizontal" heart; is your heart more vertical or more horizontal? (20 points).
4. When you raised your arm or clenched a muscle, your ECG waveform probably became overwhelmed with noise.
    a. Any explanations for why that might be? (5 points)
    b. Given that your body has many involuntary muscles that are much closer to our electrodes than year heart is, it's not obvious why an ECG doesn't pick up signals that look a lot more like an EMG! Can you come up with any ideas for why? (10 points). Consider that:
        i. Your heart has many, many cells firing synchronously. Compare this to what you learned from your sEMG lab.
        ii. A bit more physics than we will do in this course would predict that the amplitude of the voltage we read at the skin surface from a neuron or cardiomyocyte firing is proportional to the area under the action-potential curve.
    c. You probably noted that clenching your left biceps affected only Lead I, while clenching your right biceps affected both Lead I and Lead II. Why is that? (5 points)

What to turn in

Turn in your pictures and discussion questions as per above.

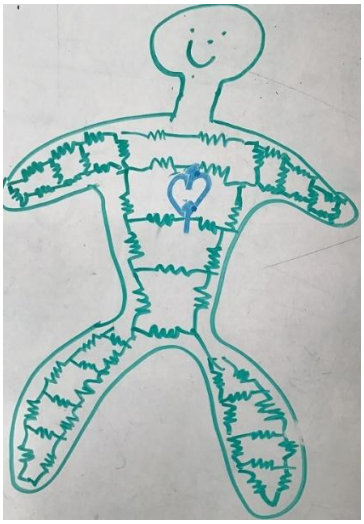
Extra credit

It turns out that attaching an electrode near your wrist gives almost identical results to our upper-arm location. However, the reasons for this are not always intuitive. In fact, it's not even obvious why sticking electrodes in your legs and arms would pick up signals that

originate in your heart in the first place! The answer to both these questions comes from better understanding the physics of electrocardiograms.

A simple but reasonable model of the heart replaces all the cardiomyocytes firing at any given time by their single summed action potential. This looks like a *current dipole* – two equal and opposite current sources a short distance away from each other. At any time, one supplies current into the body and the other sinks the returning current.

The body can be modeled as a big network of reasonably-conductive material, which we can break into a large grid of resistors:



With all due apologies for CardioMan's total lack of artistic merit, the two barely-visible blue arrows represent the current dipole. Given this model, can you explain why there is so little voltage drop between his left shoulder and his left wrist?

If CardioMan were to clasp his left hand with his right hand, would there still be almost no voltage drop between his left shoulder and his left wrist?